

3 PCR Product check with agarose gel electrophoresis

Approximate Time: 2 hours

Material

Equipment and tools

- Electrophoresis box with power supply
- Gel casting tray and comb
- Microwave-safe flask or beaker
- UV transilluminator or gel imaging system
- Balance
- Microwave
- 20µl pipette
- Camera or smartphone

Consumables

- Gloves
- Paper towels
- Parafilm
- 20µl tips

Reagents

- 1x TAE buffer
- Agarose powder
- DNA stain such as GelRed
- Molecular weight marker (DNA ladder)
- DNA loading dye
- 5 µl PCR product

Instructions

My notes

- ☐ Weigh 1g agarose for 100ml TAE buffer
- ☐ Add the agarose powder to the buffer in a microwave-safe flask or beaker.
- ☐ Microwave the solution in short bursts, swirling occasionally, until the agarose is completely dissolved and the solution is clear. Be careful not to overheat.
- ☐ Allow the agarose solution to cool slightly, then add the DNA stain. For GelRed, add 10 µL of the 10,000X stock per 100 mL.
- ☐ Gently swirl the solution to mix in the stain.
- ☐ Pour the agarose solution into a gel casting tray, inserting the comb to create sample wells. Allow the gel to solidify for 20-30 minutes.
- ☐ Once solidified, carefully remove the comb and place the gel into the electrophoresis box. Add 1X TAE buffer until the gel is submerged.
- ☐ Load 6 µL of the molecular weight ladder into the first well.
- ☐ Mix 5 µL of one randomly selected PCR sample with 1-2 µL of DNA loading dye, then load the samples into the remaining wells. Include one positive and one negative control per gel.

- ☐ Connect the electrophoresis box to a power supply and run the gel at 120 V for 30 minutes.
- ☐ Visualize the DNA bands using a UV transilluminator.
- ☐ Take a picture of the gel for documentation. Confirm the correct PCR product size.
- ☐ Repeat the gel for other samples of a stripe in case the lane was blank.

4 PCR Clean-up

Approximate Time: 1.5 hour

Material

Equipment and tools

- Hula Mixer (gentle rotator mixer)
- Vortex mixer
- Microfuge
- Timer
- Quantus® fluorometer
- Magnetic stand suitable for 1.5 ml reaction tubes
- Multichannel pipette
- 1000µl pipette
- 200µl pipette
- 20µl pipette
- 2.5µl pipette

Consumables

- gloves
- 1000µl tips
- 200µl tips
- 20µl tips
- 10µl filter tips
- 1.5ml reaction tubes

Reagents

- CleanPCR® magnetic beads
- 75% EtOH, freshly prepared from absolute ethanol e.g. 750µl EtOH with 250µl nuclease free H_2O in a 1.5ml tube.
- nuclease free H_2O
- Lambda DNA
- QuantiFluor® dsDNA dye

Instructions

My notes

- ☐ Use the multichannel pipette to pool 10µl from each reaction of all stripes in one new stripe. This is done across all 64 samples of the group.
- ☐ Pool all 8 tubes except the positive control in a 1.5ml tube and vortex 20s.
- ☐ Transfer 20µl in a new 1.5ml reaction tube for each participant.

- ☐ Vortex CleanPCR magnetic beads and shake down before opening the lid.
- ☐ Add 36µl CleanPCR® to the 20µl PCR mix.

Table 3: PCR reaction volumes and corresponding CleanPCR® volumes

PCR Reaction Volume (µL)	CleanPCR (µL)
10	18
20	36
40	72
50	90

- ☐ Incubate on the Hula Mixer for 5 minutes. Spin down.
- ☐ Place the plate on a magnetic separation device to magnetize the CleanPCR particles. Incubate at room temperature until the CleanPCR is completely cleared from solution.
- ☐ Aspirate and discard the cleared supernatant. Do not disturb the CleanPCR particles.
- ☐ Add 200µl 75% ethanol to each well.
- ☐ Incubate at room temperature for 1 minute. It is not necessary to resuspend the CleanPCR particles.
- ☐ Aspirate and discard the cleared supernatant. Do not disturb the CleanPCR.
- ☐ Repeat Steps 9-11 for a second 75% ethanol wash step.
- ☐ After the second wash step spin down the empty tube with the beads, put back on the magnetic stand and use a 10µl pipette to discard the remaining supernatant.
- ☐ Leave the plate on the magnetic separation device for 10-15 minutes to air dry the CleanPCR particles. Remove any residue liquid with a pipet.
Note: It is important to dry the CleanPCR particles before elution. Residual ethanol may interfere with downstream applications.
- ☐ Remove the tube from the magnet.
- ☐ Add 30µl nuclease free H_2O to the tube.
- ☐ Pipet up and down 20 times or vortex for 30 seconds.
- ☐ Incubate at room temperature for 2-3 minutes.
- ☐ Place the tube on the magnet and incubate at room temperature until the beads are completely cleared from the solution.
- ☐ Transfer the cleared supernatant containing purified DNA to a new 1.5ml tube.

5 Quantification

Approximate Time: 30min

Material

Equipment and tools

- Vortex mixer

- Microfuge
- Quantus[®] fluorometer
- 200µl pipette
- 2.5µl pipette

Consumables

- Gloves
- 200µl tips
- 10µl filter tips
- 0.5ml clear reaction tube (e.g. Promega[®] No. E4941, E4942)

Reagents

- Lambda DNA standard (400ng/µl)
- QuantiFluor[®] dsDNA dye
- 1µl unknown sample

Instructions

My notes

- ☐ If needed, calibrate the Fluorometer by reading blank and standard as described by the manufacturer.
- ☐ Add 200µl of QuantiFluor ONE dsDNA Dye[®] to the 0.5ml PCR tube. Protect tube from light!
- ☐ Add 1µl of unknown samples to 200µl of QuantiFluor[®] ONE dsDNA Dye in 0.5ml PCR tubes. Vortex well, and protect tube from light.
- ☐ Select the ONE DNA protocol on the Quantus[®] Fluorometer.
- ☐ Enter the volume of the unknown sample (1µl) and desired concentration units.
- ☐ Measure fluorescence of the unknown sample and record the final sample concentration results.
- ☐ Calculate the volume for 100fmol. You may use your smartphone with the QR-code below to convert the weight [ng] into molar quantity [fmol]. The COI amplicon with 9bp indexed primer has a size of 726bp, (0.73kb) or after ligation of the two adaptors 800bp (0.8kb).



Figure 2: Scan with your smartphone to visit the molecular weight to molar quantity calculator at https://www.bioline.com/media/calculator/01_07.html